

Unlocking the nexus potential: A techno-economic analysis of joint deployment of minigrids with smallholder irrigation

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ABSTRACT

Sub-Saharan Africa (SSA) has the lowest rate of electricity access and irrigation in the world, at 45% and 5% respectively. The problem is in part due to the low economic viability of rural electrification, particularly through grid extension, lack of low-cost irrigation energy, and limited access to finance for the high cost of irrigation infrastructure. Despite their promising potential, minigrids' deployment has been rather slow in SSA, partly due to the lack of viable business models for the dispersed low-income rural population. In this paper, we present an integrated model that couples a minigrid model with a crop model to explore the nexus potential of joint deployment of minigrids with smallholder irrigation to alleviate barriers to both minigrid scale-up and irrigation adoption. We apply the proposed model to three rural case studies in different geographical contexts in SSA. Our simulation results show the potential of irrigation loads to lower the unit cost of minigrid energy (LCOE) by up to 47%, depending on the quantity of irrigable land and its spatial distribution. A further scenario-based sensitivity analysis of the minigrid with and without aggregate irrigation load shows wide variations in LCOE—between a minimum of \$0.24/kWh and a maximum of \$1.2/kWh—with changes in the cost of capital and technology cost. Regarding irrigation viability, our model results show that irrigation profitability is a function of a complex interplay of factors, including crop type, soil fertility, infrastructure costs, energy prices, fertiliser application rate, and crop farm gate prices. We further test the effects of pay-as-you-go (PAYGo) business model incentives on the profitability of irrigation with minigrids; our results show that various incentives that alleviate either all or a fraction of upfront capital cost do not only mitigate irrigation adoption barrier but also enhances its profitability on the part of the farmer. Finally, our model results underscore that the economic viability of joint deployment of minigrids with smallholder irrigation depends on the optimal combination of agro-ecological factors of production (location suitability), type of crop (on a descending order, onion, tomato, and potato are the most profitable crops in our model), fertiliser application rate (near-optimal to optimal soil fertility is recommended) and cost scenarios (low cost of capital and technology preferred). In the main, our model and the results thereof highlight the potential benefits of co-planning minigrids and irrigation projects. Our model and results presented in this paper can be used as high-level decision-making pointers on the nexus planning of minigrids with smallholder irrigation farming.

Introduction

The goal of universal electricity access has spurred concerted efforts to address energy poverty in Sub-Saharan Africa (SSA). Despite this, the region still lags significantly behind the rest of the world. Approximately 65% of the SSA population remained without electricity access as of 2022 (International Energy Agency, 2022), with the rural population disproportionately underserved at a 28% access rate (Kyriakarakos, Balafoutis, & Bochtis, 2020). According to the Energy Sector Management Assistance Programme (ESMAP), due to current divergent

population growth projections and low electrification rates, an estimated 650 million people in SSA will live without electricity access by 2030 (Energy Sector Management Assistance Program (ESMAP), 2022). It is worth noting that this projected population who will live without electricity in 2030 is higher than the present rate (as of 2022), which is about 570 millions (Energy Sector Management Assistance Program (ESMAP), 2022; International Energy Agency, 2022). This projection suggests that the SSA region's population is growing at a higher rate than the growth in electrification rate. The region is largely rural,

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with more than 55% of its population residing in dispersed settlements characterised by low population density (UN-Habitat, 2020), limited income, and low potential electricity demand (Agenbroad, Carlin, Ernst, & Doig, 2018). Electrification of such a demographic through grid extension is cost-prohibitive from a system cost perspective (Bonan, Pareglio, & Tavoni, 2017). This underscores that achieving United Nations Sustainable Development Goal 7 (SDG-7), which aims to provide affordable, reliable, sustainable and modern energy for all by 2030, will require the nations of SSA to think of alternative ways, beyond centralised grids. Contemporary research literature suggests that a large proportion of the currently unelectrified rural population in developing nations would be served at least cost by minigrids, otherwise community-level microgrids (Blum, Wakeling, & Schmidt, 2013). According to Daly and Walton (2017), minigrids play a critical role in the 'universal electricity access for all' scenario, accounting for 48% of the required investments by 2030, which is corroborated in Energy Sector Management Assistance Program (ESMAP) (2019). However, despite their high prospects, their deployment scale has been rather slow in SSA as compared to other developing regions of the world, such as South Asia. The low electricity demand of the predominantly rural population of SSA (ENEA Consulting, 2020; Peters, Sievert, & Toman, 2019) and the high non-payment risks due to limited formal income (Agenbroad et al., 2018) are significant barriers to private sector investment in the mini-grid sector in SSA (Agenbroad et al., 2018; Bonan et al., 2017; ENEA Consulting, 2020; Peters et al., 2019; Williams, Jaramillo, & Taneja, 2018). Mitigating these minigrid deployment barriers calls for an integrated model that stimulates electricity demand while at the same time enhancing the economic well-being of the end-users. Admittedly, electricity demand is a function of varying contextual factors including the ability to purchase and own electrical appliances and the ability to pay for electricity bills (Agenbroad et al., 2018). Moreover, there exists a meaningful relationship between household economic poverty and energy poverty: economically poorer households rank high on the energy poverty scale (Bonan et al., 2017). To this end, reducing economic poverty has direct potential to mitigate multiple energy access barriers including low demand and non-payment risks (Bonan et al., 2017).

Evidence in the current literature suggests that productive use of electricity (PUE), using electricity for economically productive activities, including value addition in agriculture, can mitigate low demand (Agenbroad et al., 2018; Dagnachew, Choi, & Falchetta, 2023; Energy Sector Management Assistance Program (ESMAP), 2022; Kyriakarakos et al., 2020; McCall & Santana, 2018) and non-payment risks and consequently improve the viability of operating minigrids in SSA (Blodgett, Moder, Kickham, & Leaf, 2016; Bonan et al., 2017; Cabraal, Barnes, & Agarwal, 2005). According to Ngowi, Bångens, and Ahlgren (2019), PUE in agricultural activities, such as milling and water pumping, has the potential to increase the minigrids' load factor resulting in reduced levelised cost of energy (LCOE). In Kyriakarakos et al. (2020), the authors present a minigrid deployment model anchored on maize milling using a case study in Rwanda. The results of the study show that maize milling can lower the minigrid's LCOE and through cross-subsidisation, enhance access to electricity by residential consumers who will otherwise not afford the high cost of minigrid energy. These findings are corroborated in McCall and Santana (2018). Anchoring a minigrid with a fish freezing load increased its load factor and provided a revenue stream to sustain its economic operation. These examples underscore the potential of PUE in increasing electricity demand and improving affordability (by lowering the cost per unit of energy), leading to greater degree of inclusion. Despite the technical proof of concept, the deployment of minigrids in SSA has been rather slow due to the scarcity of significant PUE loads to serve as anchors (Agenbroad et al., 2018; ENEA Consulting, 2020), and therefore it is necessary to explore demand stimulation strategies (Agenbroad et al., 2018) to increase electricity demand for viable operation of minigrids and mitigate non-payment risks.

Due to the critical importance of agriculture in SSA, PUE in agriculture has the potential to realise the broadest economic dividends in terms of alleviating rural poverty and enhancing economic growth when compared to other economic sectors. According to Shah, Verma, and Pavelic (2013), agriculture creates more than 70% of informal rural employment and contributes up to 40% of the gross domestic product (GDP) of the region, despite its low mechanisation status, and has a potential to reduce poverty three to four times compared to other productive sectors (Christiansen, Demery, & Kuhl, 2011). Food production is dominated by smallholders who rely mainly on natural rainfall (Burney & Naylor, 2012); only about 5% of the region's arable land was under some form of irrigation by 2016 (International Renewable Energy Agency (IRENA), 2016), which paled compared to 40% in South Asia (International Renewable Energy Agency (IRENA), 2016). The region's agricultural productivity of maize, a staple in many countries in SSA, averages 2.0 ton/ha, which is 39% of the world's average production (Ba, 2017). This low yield is partly due to over-reliance on unpredictable rainfall, which is projected to become more uncertain in mid-to-long term futures with the ongoing climate change (Burney, Naylor, & Postel, 2013).

There is a consensus in the current literature that smallholder irrigation can increase agricultural productivity (Babovic, Milic, & Radojevic, 2009; Burney & Naylor, 2012; Hagos, Makombe, Namara, & Awulachew, 2009), reduce the current food insecurity and improve economic well-being of people in SSA (Burney & Naylor, 2012; Burney et al., 2013). According to Hagos et al. (2009), pump-based irrigation tripled tomato and onion yield and doubled cereals (maize and teff) in Ethiopia and Ghana. However, its adoption is hampered by the high cost of irrigation infrastructure (Lebdi, 2016), and lack of low cost irrigation energy (Kumar, 2005; Villholth, 2013) due to limited grid access in rural areas. Access to low-cost irrigation energy through government subsidies was instrumental to the success of the Asian green revolution, particularly in India, characterised by increased uptake of groundwater-fed irrigation and increased agricultural productivity (Kumar, 2005). From the Indian green revolution, we infer the potential of low-cost irrigation energy in accelerating the adoption of irrigation in SSA. For energy planners, policy-makers and minigrid developers, irrigation can provide the required load demand for viable deployment and operation of minigrids (Falchetta, 2021; McCall & Santana, 2018) which are currently not economically viable without subsidies (Maqelepo, Williams, & Taneja, 2022). Conversely, electrification of smallholder irrigation farming has potential to increase agricultural productivity and enhance food security of the region (Babovic et al., 2009; Nilsson, Mentis, Korkovelos, & Otswani, 2021).

However, it must be remarked here that exploitation of the nexus benefits of joint deployment of minigrids with smallholder irrigation is complex as observed in Falchetta (2021). Success of such a nexus requires policy and regulatory reforms and strategic cross-sectoral partnerships of all the involved actors—including relevant government institutions, technology providers, minigrid developers, and financial institutions. It is not until recently that some governments in SSA have started exploring the synergistic potential of interaction between agriculture and rural electrification and the need for a systems approach to the rollout of the nexus projects. A recent relevant example is the Distributed Renewable Energy–Agriculture Modalities (DREAM) project in Ethiopia (Energy Market Accelerator Ethiopia (AMAE), 2021). The project aims to deploy irrigation-anchored minigrids as off-grid electrification vehicles in Ethiopia. In that perspective, irrigation loads will provide stable and predictable load demand to mitigate the low demand barrier, and, conversely, the minigrids will provide low-cost energy for smallholder irrigation.

Besides the few recent pilot projects like the DREAM project (being piloted as from 2022) described above, there is scant evidence in the contemporary literature on minigrid deployment with irrigation in SSA and most importantly, literature on detailed economic analysis of the nexus potential to inform conditions for profitable operation of

minigrids with smallholder irrigation is lacking. Where electrification and irrigation are jointly studied in the current literature, the focus has mainly been on assessing the latent agricultural energy demand with minimal effort in quantifying the impact of such loads on minigrids (Falchetta et al., 2021; Izar-Tenorio, Jaramillo, & Williams, 2021; Nilsson et al., 2021), otherwise, the economic benefits of irrigation to the farmer have not been analysed and quantified (Shoeb & Shafullah, 2018; Uamusse, Tussupova, Persson, & Berndtsson, 2019). The results of Falchetta et al. (2021), Izar-Tenorio et al. (2021) and Nilsson et al. (2021) show that irrigation require substantial amounts of energy that can improve the financial health of utilities in the region if integrated in electrification planning. In Juturu et al. (2023), the authors under-take a scenario-based analysis of the impacts of groundwater-fed irrigation (GWI) demand on centralised grid demand in Ethiopia. Their results show a significant increase in generation capacity factor and grid capacity requirement with GWI integration. However, no effort is made in the analysis to assess the economic impact of GWI to the utility and agricultural productivity. In Wamalwa, Izar-Tenorio, and Williams (2022), potential economic benefits of irrigation farming of select crops in Kenya are analysed but the impact of irrigation on electrical utilities is not evaluated. Where powering of the community and irrigation loads from the same minigrid has been studied, the economic impacts of irrigation to the farmer have not been considered (Shoeb & Shafullah, 2018; Uamusse et al., 2019). For instance, in Uamusse et al. (2019), the authors study the performance of a hydro minigrid with an array of PUE activities including milling of cereals and water pumping for residential use in a rural village in Mozambique. Their findings show that PUE loads can lower the LCOE of the minigrid. However, as with all the other literature studied above, Uamusse et al. (2019) does not consider the analysis of the economic benefits of irrigation for the farmer.

In this work, we integrate a biophysical crop model with a minigrid model to study the nexus benefits of joint deployment of minigrids with smallholder irrigation farming. We employ the proposed model to hypothetical case studies with a goal of understanding conditions under which, when operated together, both minigrids and smallholders can both realise positive benefits. We assess the benefits of irrigation loads on the minigrids in terms of changes in LCOE, a reduction in LCOE being a positive performance indicator. Besides the impacts of irrigation loads on minigrids, we also investigate cost-competitiveness of minigrid-powered electric pumping over dispersed diesel engine and solar photovoltaic (SPV) pumping, as the latter two are immediate alternatives in unelectrified rural communities (Falchetta et al., 2023; Xie, Ringler, & Mondal, 2021). We further assess the effects of pay-as-you-go (PAYGo) innovative incentives to irrigation adoption and profitability on the part of the farmer. To this end, we answer the following research questions in our work: (i) What is the impact of smallholder irrigation loads on the LCOE of the solar PV-battery-diesel minigrid as a technology option for off-grid electrification in SSA?; (ii) What is the impact of smallholder irrigation on the profitability of the commonly grown crops in SSA?; (iii), what is the effects of PAYGo innovative business model on irrigation adoption and profitability?; (iv) Under what conditions do minigrid operators and smallholder farmers see positive benefits when co-planned together?

The contributions of our work are as follows: (i) We propose an integrated model to study the nexus benefits of co-planning minigrids with smallholder irrigation farming; (ii) we assess economic benefits of smallholder irrigation of a range of commonly grown crops in SSA under uncertain production conditions and cost scenarios to determine the conditions under which it is profitable to co-deploy minigrids with smallholder irrigation loads; (iii) The high upfront capital cost being a significant barrier to irrigation uptake in SSA (Lebdi, 2016), in this paper, we propose an innovative PAYGo business model to amortise capital costs over the project life for the farmer as in Falchetta (2021) rather than facing it upfront. The remainder of the paper is organised as follows: Section “Model development and case study selection” lays out the conceptual framework and the methods employed, including

the data sources used in our analysis; Section “Results and Discussion” presents the model results; Section “Sensitivity analysis” presents sensitivity analyses of the model performance and results; Section “Policy implications” presents policy recommendations emerging from our study, while Section “Conclusion” concludes the paper.

Model development and case study selection

Preliminary spatial analysis and case study selection

Viability of a location for deploying an irrigation-anchored minigrid depends on a number of factors including availability and viability of alternative irrigation energy sources (dispersed off-grid technologies and grid proximity in this case), availability of irrigable cropland, and population. The distances from the centralised grid beyond which it is considered viable for minigrid electrification in the three countries are shown in Table 1. The table also contains other constraints for location viability for the nexus minigrids-irrigation projects; cropland, and population. We rely on the gridded Population of the World (GPW), gridded gross domestic product (GDP), and cropland data to identify viable locations for the case study; a summary is given in Table SI 4 in the supplement. Both the GPW and GDP data are provided at 30-arc-second (approx 1 km at the equator) spatial resolution while the cropland dataset is provided at 5-arc-min (approx 10 km at the equator) resolution. Therefore, the first step entails re-sampling of the GPW and GDP data to the cropland resolution (5-arc-min) before further geospatial analysis to determine viable locations (5-arc-min gridded cells in this case) for minigrid deployment. The resultant spatial maps for the three countries (Zambia, Ethiopia and Nigeria) are shown in Section SI 2.1 in the supplement. We have considered a 5-arc-min gridded cell as a scale for the minigrid-irrigation nexus project, as in Farthing et al. (2023), if the cell meets the criteria defined in Table 1. As a general remark, we note that minigrids can be deployed in locations with even less than 1200 people (in Ethiopia the minimum population is 250 people (Holzigel, 2021)), however, in this paper we consider a minimum population of 1200 to achieve a minimum of 300 households based on the household size data used in the model.

Table 2 describes the identified case studies used to test our minigrid deployment model. We estimate the GDP per capita at purchasing power parity, GDP (PPP), at the three case study locations, by dividing the obtained GDP (PPP) values in each 5-arc-min cell by the cell population. For depth-to-groundwater table (DTW) is a 5-arc-min spatially resolved data from the British Geological Survey (BGS) (MacDonald, Bonsor, Dochartaigh, & Taylor, 2012) while daily solar insolation (GHI) is obtained from NASAPOWER (Stackhouse, 2022), averaged over the 20-year period (2001–2020).

The case study locations are 5-arc-min gridded cells identified as ‘site 1’ of the five marked sites in each country (Figures SI 4, SI 5, and SI 6 in the supplement); that is Northern Zambia, North-Western Ethiopia, and Northern Nigeria. All the three case study sites are off-grid agricultural communities that can be electrified via minigrids; In Nigeria, we have considered a location in the local government of Zuru in Kebbi State. In Zambia, we have considered a location in the Northern province of Luapua in Nchelenge district while in Ethiopia, we have considered a site in the Awi zone. Before characterising the three case studies, it is important to remark here that the selection of the three sites was random and as such, any one of the five sites marked in red colour or the remainder of the potential minigrid (MG) locations shown in green colours in Figures SI 4, SI 5 and SI 6 can be used in the study. The choice of the three countries is informed by the high concentration of rain-fed smallholder cropland, large populations without electricity access (mainly Ethiopia and Nigeria), and geographical locations within the SSA sub-regions (Southern, Eastern and West Africa).

Table 1
Criteria and input data used in formulating land exclusion constraints.

Data	Inclusion criteria	Data source
Cropland	>75 ha	IFPRI's MaPSPAM (International Food Policy Research Institute, 2020)
Population	>1200	Gridded Population of the World (GPWv4) (Center for International Earth Science Information Network - CIESIN - Columbia University, 2016)
Distribution grid Nigeria	>25 km (International Finance Corporation (IFC), 2018)	African Electricity Grids Explorer (The World Bank Group, 2018)
Distribution grid Zambia	>15 km (Farthing et al., 2023)	African Electricity Grids Explorer (The World Bank Group, 2018)
Distribution grid Ethiopia	>25 km (Holzigel, 2021)	African Electricity Grids Explorer (The World Bank Group, 2018)

Table 2

A summary of the spatial features of the study locations (100km² spatial cells). The depth-to-water table (DTW), is given as a range over a 5 km spatial resolution. GHI is the average daily global horizontal irradiation. The capita GDP is based on the 2010 purchasing power parity of the dollar with respect to the local currencies. HH size is the national averages of the rural household sizes.

Site 1	Lat (deg)	Lon (deg)	DTW (m)	GHI (kWh/m ²)	Cropland (ha)	Population	GDP (\$/per person)	HH size (Mentis et al., 2017)
Ethiopia	10.875	36.375	7–25	6.10	5398	2135	460	5.2
Nigeria	11.4583	5.375	7–25	5.39	3677	2471	387	5.3
Zambia	−9.1825	28.957	7–25	5.61	394	1384	758	4.6

Farmland distribution and minigrid network costs

The minigrid network and its cost depends on electricity demand, the number of connections and the spatial distribution of those connections ([Ciller, Lumbreras, & González-García, 2021](#)). From [Table 2](#), We estimate a minimum of 300 residential households in each of the three sites, for which a 300 household baseline minigrid is sized, assuming 100% subscription. Besides the electricity demand for the hypothetical 300 residential households, we also include electricity demand for productive and social uses such as schools, rural clinics, domestic water pumps and street lights as in [Farthing et al. \(2023\)](#). See a detailed account of community load profile estimation in Section SI 2.1.1 in the supplement. In SSA, it is common to find house-holds in close proximity to one another clustered around a community core with farmlands located a few kilometres away from the community core. This implies a high distribution network cost for the minigrid that powers both communal and irrigation loads. To account for the impacts of farmland locations on the minigrid network costs, we consider the four hypothetical scenarios: cases C-I, C-II, C-III, and C-IV. Case C-I is a baseline minigrid deployed to serve the community hourly load while cases C-II to C-IV consider inclusion of irrigation loads in C-I minigrid. Case C-II has a moderate irrigation load (30 ha) located 1–2 km away from the community core; case C-III assumes C-II with an additional 30 ha irrigable farmland located 2–4 km from the community core; case C-IV assumes C-III with an additional 15 ha of farmland located 4–5 km from the community core. A detailed account of these scenarios is presented in Section SI 3.1 in the supplement. The differences in the distances of the locations of the farmlands away from the community core (which we assume is also the minigrid generation hub) implies that each of the cases C-II to C-IV has different medium voltage (MV) and low voltage (LV) network costs associated with it. For instance, beyond 1 km radius from the minigrid generation hub, the network voltage drop may exceed 10% of the nominal values ([Suryani, Fadhillah, Saichu, & Mubarak, 2021](#)) thus an MV network (including two service transformers—one for stepping up and one for stepping down voltages) is required to ensure that power of acceptable voltage levels is delivered to the farmland equipment (electric pumps in this case).

As a general remark, irrigation land size of 30 ha, 60 ha, and 75 ha used in C-II, C-III, and C-IV minigrid deployment scenarios are not necessarily an optimal consideration for a single minigrid. The land sizes are chosen to generate moderate irrigation demand to study its impact on the minigrid energy cost. Optimal irrigation load on the minigrid depends on several factors such as the irrigation land availability and its spatial distribution, and grid infrastructure design constraints. For example, the DREAM project in Ethiopia ([Energy Market Accelerator Ethiopia \(AMAE\), 2021](#)) is deploying nine (as of September 2022) minigrids anchored by smallholder irrigation in nine villages with an

average of 325 (178–652) households and 289 ha (42–473 ha) of irrigable farmlands ([Veritas Consulting, 2020](#)). To this end, our model and the results thereof will only be high level assessment and considerations for co-planning minigrids with irrigation but optimal design and results for the considered locations; the latter will depend on the available irrigation land at a study location. We test our model on a range of select crops including maize, wheat, beans, potato, onion, and tomato. The crops are selected based on their relative importance in SSA and more so, in the three case study countries. Maize, wheat, bean, and potato are common staples grown in SSA while onion and tomato are high value horticultural crops mainly grown as cash crops. Respective runs of the rainy and dry seasons used to inform planting dates in the model are given in [Table SI 6](#) in the supplement.

Methods, data and their sources

In this section, we describe the methods used in our model. They include the methods for estimating both community and irrigation load profiles, sizing of the minigrid components, calculation of the minigrids' LCOE, and the net present value (NPV) of irrigation as a viability metric for irrigation. [Fig. 1](#) shows the schematic layout of our modelling framework. The core model comprises of two modules; the biophysical crop module and the minigrid module. We develop a simplified biophysical crop model based on the Food and Agricultural Organisation's (FAO's) Penman–Monteith equation ([Allan, Pereira, & Smith, 1998](#)) to simulate irrigation water requirement, W_{irr} , used for sizing of irrigation infrastructure and minigrid components for Cases C-II, C-III, and C-IV minigrids (see section "Preliminary spatial analysis and case study selection").

For irrigation yield gains, we follow the methods described in [Izar-Tenorio et al. \(2021\)](#) to simulate crop yields over a 20 year period (from 2001 to 2020) in the FAO's AquaCrop ([Vanuytrecht et al., 2014](#)) model under both rainfed and irrigated conditions in rainy and dry seasons (see [Table SI 6](#)) and then calculate the yield gain in both rainy and dry seasons as the difference between irrigated yield and rainfed yield assuming other production factors and conditions remain the same. The annual yield gains used in the farmers' economic module are sums of yield gains in the rainy and dry seasons of each of the case study locations. Access to farm inputs, including fertiliser and improved seed cultivars is a major problem for smallholder farmers in SSA ([Fuglie & Rada, 2013; Jayne, Mather, & Mghenyi, 2010](#)). To ensure that simulated yields are representative of common practices in the SSA region, we use crop files calibrated with parameters of traditional low-yield cultivars in AquaCrop model as in [Izar-Tenorio et al. \(2021\)](#); a detailed account is given in Section SI 2.2 of the supplement.

To assess the impact of soil fertility on yield, we conduct a sensitivity analysis over a ranger of fertiliser application rates including

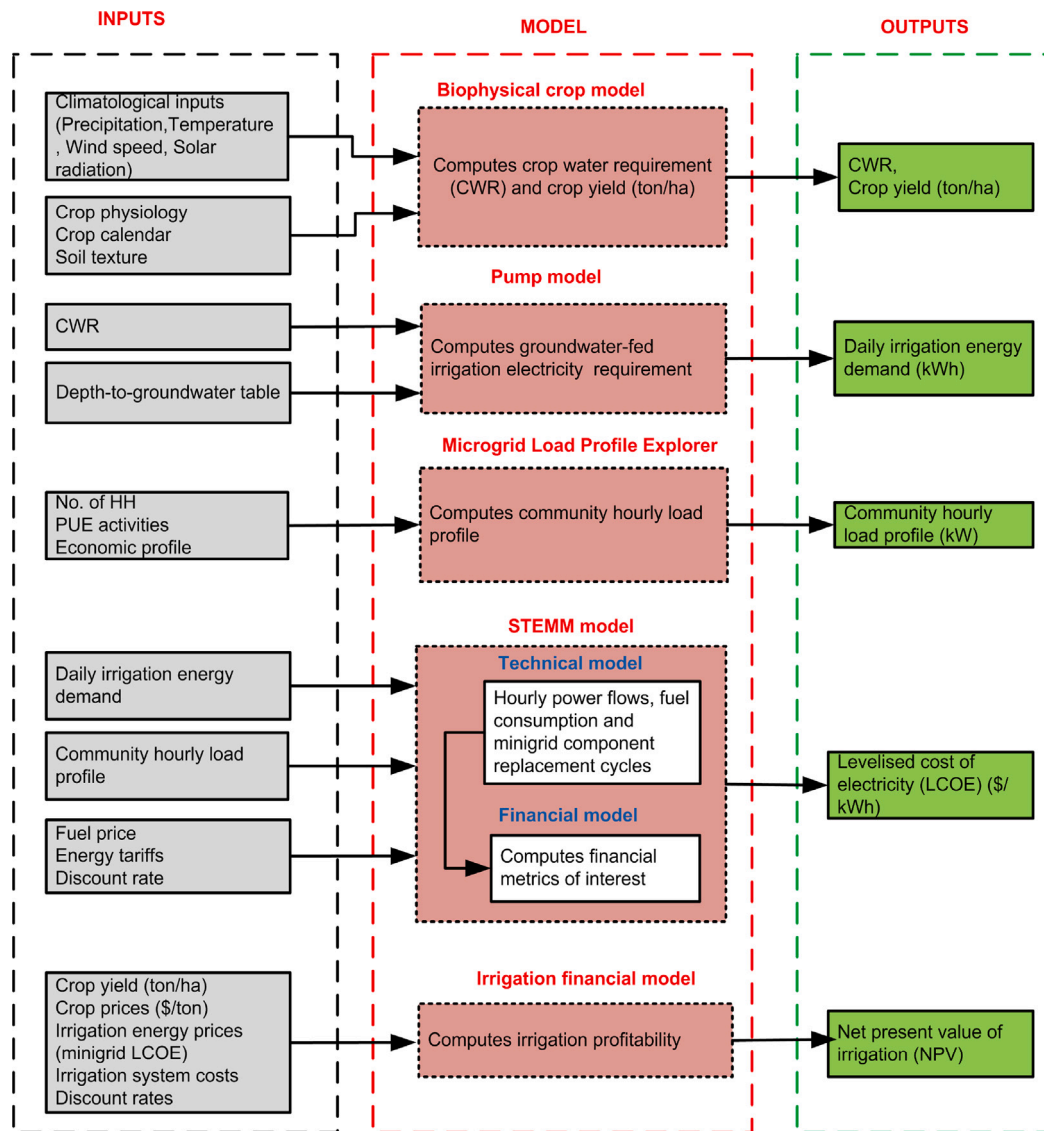


Fig. 1. Schematic layout of the model workflow. The STEMM model was originally developed by Williams et al. in Williams et al. (2018) to simulate techno-economic operation of minigrids in SSA from an investor perspective. In this work, the output of STEMM model is the minigrid LCOE while the output of irrigation financial model is the net present value (NPV) of cashflows of irrigation. HH = number of house-hold, PUE = Productive use of electricity.

optimal (100% required rate), near-optimal (85% of the required rate), moderate (50% of the required rate), and low soil fertility (25% of the required rate) as in Izar-Tenorio et al. (2021). The crop and minigrid modules are inter-linked with a pump module for estimating irrigation energy requirement using the pump head and W_{irr} as inputs. Key input data to the crop and pump modules are given in Table SI 7 in the supplement.

In this paper, we have used climate data averaged over a 20-year period, between 2001 and 2020, to capture seasonal and inter-annual variability in W_{irr} due to variability in climate. To estimate communal electricity demand, we rely on the Microgrid Load Profile Explorer Tool (MLPET) developed by the National Renewable Energy Laboratory (NREL) for estimating hourly annual load profiles for minigrids in SSA (Li, Salasovich, & Reber, 2018). The output of the pump module, which is the daily irrigation energy demand, E_{irr} , and the communal hourly load profile are fed into the technical module of STEMM. The output of STEMM is the minigrid metric of interest, which in this case is the LCOE. Finally, the irrigation financial module takes irrigation yield gain from the crop module, crop and fuel prices (the minigrid LCOE in this case), irrigation infrastructure development costs, and discount rates as inputs to compute the irrigation profitability for the

farmer. Since using LCOE as a minigrid tariff returns non-negative NPV for the minigrid, positive NPV of irrigation for the smallholder farmer powered by the minigrid is a positive indicator for viable condition for co-deploying minigrid with smallholders.

Key modelling assumptions

To understand the validity and limitations of our model, we describe, in this sub-section, the key assumptions considered in the modelling process.

The first assumption regards the irrigation water source. In this work, we consider groundwater and assume aquifer storage at each location is sufficient to meet irrigation demand. We consider groundwater because of its natural abundance in Africa (MacDonald et al., 2012), even spatial distribution that allow its access virtually from anywhere where drilling is technically and economically feasible, and lastly because of its slow response to meteorological conditions than surface water and, as such, it provides a natural buffer against climate variability, including drought (MacDonald et al., 2012).

The second assumption regards the cost structure of the minigrid. The range of costs used in this paper, see Table SI 10 in the supplement,

are based on a research on minigrids by the ESMAP of the World Bank Group (Energy Sector Management Assistance Program (ESMAP), 2022). We take into account the impacts of uncertainties in network costs to the model performance by undertaking parametric sensitivity analyses of the effects of farmland location from the minigrid to the distribution network cost and hence the minigrid LCOE (see Section “Farmland distribution and minigrid network costs”).

Thirdly, on irrigation infrastructure cost allocation, we assume a 10 ha cropland per borehole according to Shah, Namara, and Rajan (2020). Most farmer-led shallow/deep well small-scale irrigation projects in developing countries involve a group of farmers (or cooperatives) with up to 10 ha (with a holding range of 0.1–2 ha per farmer) of cropland distributed among themselves sharing a borehole (Shah et al., 2020). Since in our model we compute profitability per hectare of irrigated cropland, drilling and development costs of a borehole are prorated to the hectare level. To account for low voltage network costs associated with irrigation, we assume that each of these boreholes is an independent connection node in the network with its own associated network cost.

The fourth assumption is on the system boundary for the comparison of minigrid, diesel engine, and solar PV powered irrigation. In this paper, in addition to investigating the impacts of irrigation loads on the LCOE of the minigrid, we also compare the pumping costs of the three technologies mentioned above to determine the least cost option for the smallholder farmer. We simplify the modelling complexity by assuming that the system’s boundary for comparative analysis is the power unit and the pump, as shown in Figure SI 15 in the supplement; we ignore the cost of irrigation water conveyance in the comparative analysis since it is common to all three pumping systems.

Lastly, in calculating irrigation profitability using equation (C.2), we considered 15% income losses for the farmer in post-harvest handling. According to Reckefeller Foundation (2015) up to 30% of food produced in SSA is wasted or lost between production and consumption, resulting in an approximate 15% income loss for the farmer.

Irrigation energy requirement

In general, irrigation energy, E_{irr} , briefly explained here with detailed formulation in the supplement, depends on irrigation water requirement, W_{irr} , and the total dynamic head (TDH) of the pumping system. W_{irr} is the gross crop water requirement, W_c , including irrigation application losses, applied to supplement crop evapotranspiration needs, ET_c , not met by effective precipitation and the available soil water storage. TDH comprises the static head, the operating pressure, and head loss due to friction in the water conveyance system.

Selection of the right irrigation method is essential for sustainable water and energy use. In this paper, we consider pressurised irrigation systems (drip and sprinkler) as the most suitable due to their high water application efficiencies. We have considered drip irrigation for vegetable crops (onion and tomato crops) and sprinkler for the rest of the crops (beans, maize, wheat and potato). Tomato and onion crops are vulnerable to diseases when water is applied aurally. For water withdrawal efficiency, we assume an average value of 60% for all the crops considered across the three countries after Falchetta et al. (2023). For irrigation system efficiencies, we assume 70% and 85% for sprinkler and drip respectively (El-Shimy, 2013; Phocaides, 2007); a summary is given in Table SI 1 in the supplement.

The minigrid model

To simulate the minigrid operation, we rely on the STEMM model (Williams et al., 2018), which comprises of two modules; (i) the technical and (ii) financial modules. Briefly explained here with detailed formulation in the supplement, the technical module is a minigrid simulation tool that simulates the operation of a solar PV–diesel–battery minigrid. The inputs to the minigrid, include the hourly solar

Table 3

A summary of the minigrid system component sizes for the four minigrid models (Cases I, II, III, and IV).

MG item	Zambia				Nigeria				Ethiopia			
	C-I	C-II	C-III	C-IV	C-I	C-II	C-III	C-IV	C-I	C-II	C-III	C-IV
Solar PV (kWp)	27	112	150	155	30	154	176	185	26	36	45	50
Inverter (kWp)	27	112	150	155	30	154	176	185	26	36	45	50
Genset (kWp)	16	39	45	47	16	41	44	48	16	16	20	23
Battery (kWh)	90	127	130	130	90	120	140	140	90	127	142	142

Table 4

Power ratings of the irrigation pumps for 1 ha of maize cropland.

Site	Pump (kW)	Motor (kW)	Solar PV (kWp)
Ethiopia	3.8	5	4.8
Nigeria	5.2	7	6.5
Zambia	4.8	5.5	6.8

irradiation, hourly temperature, and the load profile to be served. The communal electricity demand, shown in Figure SI 7 in the supplement, is inflexible and must be met at each hour it occurs while the irrigation energy demand is modelled as a flexible daily demand that can be shifted over time within a 24 h period to minimise daily irrigation energy cost. The outputs of the minigrid, which include the hourly powerflows and diesel fuel consumption, are fed into the financial module that computes the metric of interest, which in this paper is the LCOE. Table 3 shows the system components sized based on the communal load (for Case C-I) and the aggregate communal and irrigation load profiles for Cases C-II, C-III, and C-IV minigrid models; See Figure SI 8 in the supplement for the daily seasonal irrigation electricity demand. As shown in Table 3, system components for C-I to C-IV models vary across the three locations due to variations in climate. As a general remark, in all the cases C-I to C-IV, the battery storage is sized for the residential communal load but it is available to meet the irrigation energy demand as determined technically and economically feasible by the minigrid dispatch algorithm.

For comparative analysis of the minigrid-powered electric-motor, diesel-engine, and the solar PV-powered pumping, we size the three systems based on the peak $W_{irr}(k)$ for 1.0 ha of the crop with the highest W_{irr} , which is maize in this paper, taking into consideration the engine and electric motor power conversion efficiencies shown in Table SI 11 in the supplement. The 1 ha sizing reference here aligns with an assumption in our model that each farmer in the model farms an average of 1.0 ha and therefore, an alternative to the minigrid is buying a diesel or solar pump to meet their 1 ha irrigation requirements. Pump power ratings for the three case study sites are given in Table 4.

Techno-economic model development

In this sub-section, we present formulation of the metrics used in this paper. We use LCOE, briefly explained here and formulated in the supplement, as a metric to assess the impact of irrigation loads on minigrids. In this sense, a decrease in LCOE due to the integration of irrigation loads into the minigrid is a positive performance indicator. We then use the obtained LCOE values to compute the NPV of irrigation yield gain as a profitability metric, a key condition for joint deployment of minigrids with smallholder irrigation. We use positive NPV as a proxy for the farmers’ ability to pay for the irrigation energy cost. In our model, positive NPV arises when revenue from irrigation yield gain is higher than the total cost of installing and operating the irrigation system. A detailed account of the model input parameters, and cost assumptions is presented in Section SI 4 of the supplement.

For comparative analysis of diesel engines, solar PV and minigrid-powered electric pumps, we use the levelised cost of pumping energy (LCOP) as a comparative metric. Analogous to the LCOE, we define

LCOP as the NPV of the total costs of the pumping system (comprising the power unit and the pump) divided by the NPV of the total irrigation energy delivered. The Pumping system costs include the capital, operations and maintenance, and fuel expenditures of the pump and its power unit. For instance, the pumping systems cost for a minigrid powered electric motor-pump includes the capital cost of the motor-pump system, the minigrid connection costs/fees, minigrid energy tariffs costs, and the operations and maintenance costs of the system over its lifetime.

Economic benefits of irrigation

Lack of detailed information and data on irrigation systems in SSA makes it difficult to estimate irrigation yield benefits. Where irrigation yield data exist, such as the projected yield from MapSPAM (You, Wood, & Wood-Sichra, 2009) and National yield statistics by FAO-STAT (Food and Agricultural Organisation (FAO), 2022), the data is either fragmented, and/or the information on cropping seasons and fertiliser use rates is lacking, making it difficult to estimate irrigation benefits from such a dataset. It is also worth noting that MapSPAM and FAOSTAT yield data, respectively, are historical sub-national and national point estimates. These low-spatial resolution point estimates are not meaningful where the impact of spatial and temporal variability in climate – mainly precipitation – on crop yield is of interest at a higher spatial resolution (5-arc-min) used in this paper. In our work, we mitigate the data paucity and uncertainty challenge by using potential yield estimates from a crop model (AquaCrop model of the FAO). Yield simulation is relevant in understanding the benefits of irrigation and fertiliser application rate, and thereby inform policy measures that can close yield gaps and food insecurity in SSA.

One of the major barriers to the adoption of irrigation in SSA is the high upfront cost of irrigation infrastructure (Lebdi, 2016), including the irrigation system costs, the borehole drilling and development costs for the groundwater-fed irrigation, and the cost of the pump. Many smallholder farmers cannot afford them. Therefore, a mere analysis of irrigation profitability is not enough if farmers cannot afford the upfront cost. There are various innovative business models designed to alleviate access barriers in the extant literature including loans, capital grants, fee for service, and pay-as-you-go (PAYGo) models (Bhattacharyya, 2013; Falchetta, Michoud, Hafner, & Rother, 2022; Falchetta et al., 2023). In Falchetta et al. (2023), a novel business model that amortises the capital cost (all or a fraction) of the solar irrigation system is proposed; which is a popular model in the solar home systems (SHSs) market (Mergulhao, Capra, Voglitsis, & Parikh, 2023). In this paper, we propose the following PAYGo incentives for alleviating upfront capital cost barrier and also to understand their impacts on irrigation profitability; (i) 'All costs incentive' model which amortises all the upfront capital cost of the irrigation project over its life; (ii) 'Irrigation system incentive' model which amortises only the irrigation system costs (including the pump); (iii) 'Borehole incentive' model which amortises only the borehole drilling and development costs of the project; (iv) 'Zero incentive' model, a baseline case where all the upfront capital costs are faced by the farmer. In all these scenarios, the farmer is responsible for the system depreciation costs including replacements. For instance, in 'All costs incentive', the full initial cost of the project are amortised together with the operations and maintenance costs of the project, including replacement costs, over its lifetime. A detailed account of these incentives is given in Section SI 3.4 of the supplement.

Results and discussion

Minigrid levelized cost of energy

Fig. 2 shows the minigrid LCOE at the three case study locations across the four minigrid cases (C-I to C-IV). Despite having the same communal and irrigation loads, the three locations register different

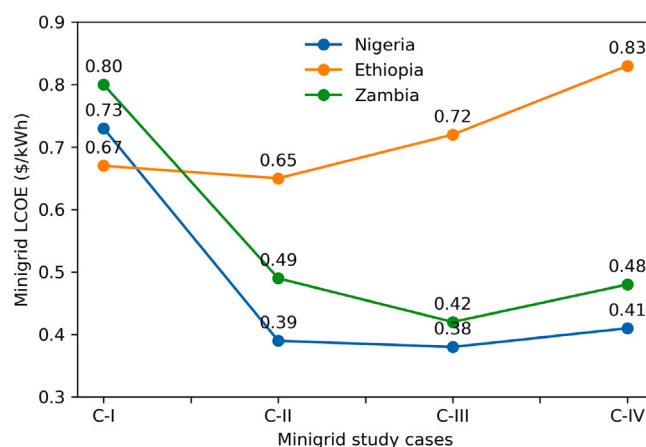


Fig. 2. The levelized cost of energy (LCOE) for the four minigrid cases (C-I, C-II, C-III, and C-IV) based on the median discount rate and median minigrid equipment costs; see Table SI 10 in the supplement.

LCOEs for each minigrid case with the site in Ethiopia consistently having higher LCOEs as compared to the sites in Zambia and Nigeria. At first, 30 ha irrigation demand results in a drastic decrease in LCOE of C-II minigrid in Nigeria and Zambia. An increase in irrigation loads in C-III and C-IV results in a slight decrease in LCOE before an increase in C-IV in Nigeria and Zambia. In Ethiopia, case C-II results in a marginal decrease before subsequent increase in C-III and C-IV. The results suggest different optimal irrigation load integration for each location if LCOE decrease is to be sustained. At the site in Ethiopia, irrigation loads beyond 30 ha results in LCOE increase while in Nigeria and Zambia is 60 ha.

The LCOE trend in Fig. 2 can be explained by the disproportionate changes in the load factors of the minigrids with seasonal irrigation loads and increases in minigrid infrastructure costs resulting from increasing peak irrigation loads and distribution network expansion to power farms away from the generation hub; see Section SI 3.1 on the effects of farmland locations on the minigrid network costs. The observed trend highlight the benefits of irrigation loads in lowering the minigrid energy costs. However, this is only possible to a certain extent as further increase in seasonal irrigation loads (in C-IV) results in a decrease in minigrid load factor and eventual increase in LCOE. Our results are consistent with the findings in the literature (Booth et al., 2018); that seasonal PUE loads can either reduce or increase LCOE of minigrids depending on operation scenarios.

Fig. 3 shows the fractional contribution of each of the components to the minigrid LCOE. As shown in the figure, distribution network costs accounts for up to 52% of the system costs (in Ethiopia); which is attributed to the infrastructure expansion to cover irrigation loads, that are up to 4 km away from the minigrid generation hub—according to the assumption in Table SI 8 in the supplement.

The LCOP values of the three pumping technologies are presented in Table 5. As shown in the table, the LCOP varies with location and technology with minigrids as the least cost pumping power source at the sites in Zambia and Nigeria; in Ethiopia solar PV pumping is slightly cheaper than minigrid and diesel pumping with the latter two having no significant difference. The Variations in LCOP are due in part to variations in system size (Table 4), irrigation energy variability (Figure SI 8 in the supplement) and fuel cost (Figure SI 16 in the supplement).

As a general remark, although solar PV pumping is slightly cheaper than minigrid-powered pumping in Ethiopia, justification for minigrids over distributed diesel and solar powered irrigation is in the fact that minigrids do not only provide motive power for irrigation; their excess capacity can be used for other productive and social uses. For instance, the excess minigrid power could be utilised for post-harvest agro-processing, and village lighting among others.

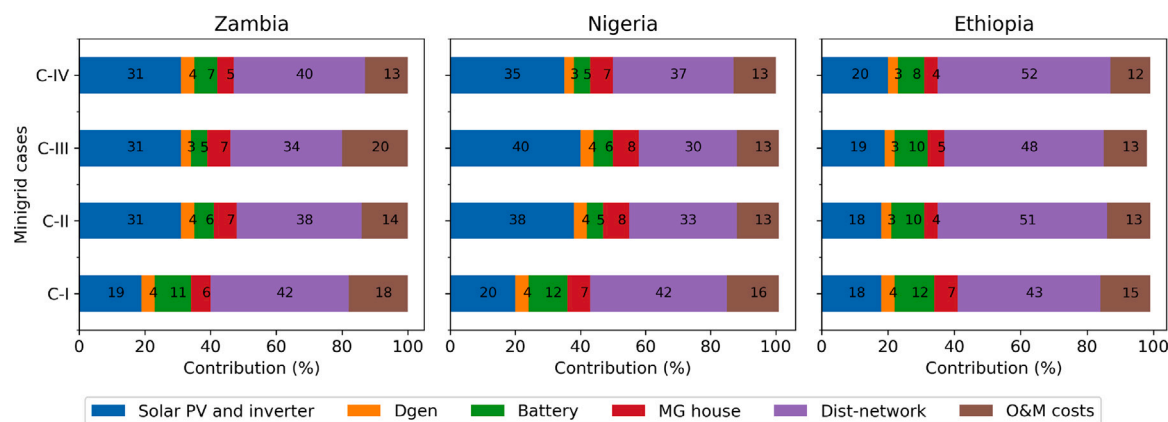


Fig. 3. LCOE cost distribution by minigrid components. The results are based on life cycle costs associated with the median LCOE values of Fig. 2. Dgen = Diesel generator.

Table 5

The levelised cost of pumping energy (LCOP) for the minigrid powered electric, solar PV, and diesel engine powered pumping. The analysis is based on 1 ha of cropland and C-II LCOE values for the minigrid.

Pumping system	Ethiopia LCOP (\$/kWh)	Nigeria LCOP (\$/kWh)	Zambia LCOP (\$/kWh)
Diesel engine pumping	0.87	1.05	1.04
MG-electric pumping	0.86	0.70	0.77
Solar PV pumping	0.83	0.95	0.91

Economic viability of irrigation

Fig. 4 shows irrigation yield gain (ton/ha) and the associated annual profitability (\$/ha/yr). The reported yield gains are sum of the yield gains in the rainy and dry season cropping averaged over a 20-year period (2000–2020); see Figure SI 10 in the supplement for the 20-year averaged rainfed and irrigated yields of the six crops. The results highlight the dependency of irrigation benefits on geographical location and the type of crop grown. Horticultural crops – onion and tomato – record the highest annual profits (up to \$1360/ha and \$1030/ha for onion and tomato respectively) while maize, bean and wheat crops are not economically viable to irrigate across the three locations except maize in Nigeria. It is profitable to irrigate potato crop in Zambia and Nigeria with the latter location recording relatively higher benefits. The differences in irrigation profitability across the three locations can be explained by the differences in the model inputs including discount rates, crop prices, irrigation infrastructure costs (see Figure SI 11) and climate (affects crop yields) which are country-specific. Our results are consistent with previous research work that smallholder (often 1–2 ha) farming of common staple crops in SSA – such as maize, beans and wheat – is not profitable (Giller et al., 2021) due to the low market value of these crops (Burney et al., 2013).

As a general remark, based on the data given in Table 2, we have estimated a minimum of 300 households in each of the case study locations for which we have designed a 300 household minigrid. From the household size data and average arable land per person in the three countries (Alliance for Green Revolution in Africa (AGRA), 2014), we have estimated an average farmland of 0.8–1.2 ha (1.0 ha) per household; see Table SI 13 in the supplement for summary statistics. Based on this assumption, of 1.0 ha/household, we can use the reported annual per hectare profitability results as equivalent household annual income from irrigation farming. To this end, comparing the results of Fig. 4 and the GDP per capita (\$/pp) of 460–758 for the three locations (Table 2) given the household size (Table SI 13 in the supplement), irrigation can significantly improve the productivity, and potentially the well-being, of the affected communities.

Sensitivity analysis

The model results reported so far are based on cost estimates from literature with high spatial and temporal variability. Minigrid and irrigation system costs change with time (International Renewable Energy Agency (IRENA), 2016) and location (country) as well as other factors, such as technology selection, logistics, and import tariffs (Energy Sector Management Assistance Program (ESMAP), 2022), which affect the minigrid LCOE. On the part of the farmer, irrigation infrastructure costs are highly variable depending on the service providers, project location, and project specifications (Villholth, 2013). Therefore, single-point estimates cannot tell the entire story about the economic viability of irrigation. To this end, on the part of the minigrid developer, we perform a sensitivity analysis to understand how changes in discount rates (DR) and technology costs (TC) affect the minigrid LCOE. We have considered three cost scenarios; low-cost, median-cost, and high-cost. The ‘low-cost’ and ‘high-cost’ scenarios combine the lower and upper bounds of TC and DR respectively. Meanwhile, the ‘median-cost’ scenario is based on the median TC and DR values. On the part of the farmer, we analyse how uncertainties in fertiliser application rates, irrigation infrastructure costs, and irrigation energy tariffs (minigrid LCOE in this case) affect irrigation profitability. This is necessary in the assumption that the farmer’s ability to pay for irrigation energy costs depends on irrigation profitability and hence viability of the nexus planning. For soil fertility, we consider four different soil fertility levels; optimal (100f), near optimal (85f), moderate (50f) and low fertility (25f) as in Izar-Tenorio et al. (2021) to study how soil fertility affects crop yield and irrigation profitability. Sensitivity analysis results are presented in the following section.

Minigrid levelized cost of energy

Fig. 5 shows the minigrid LCOE with changes in discount rates (DR) and technology costs (TC). In the figure, we have considered nine cost scenarios comprising different DR/TC combinations for each of the minigrid study cases; C-I to C-IV. As shown in the figure, changes in DR and TC result in a range of LCOE values with a minimum of \$0.24/kWh (in Nigeria) and a maximum of \$1.2/kWh (in Zambia), depending on location, minigrid cases, and TC/DR combinations. In all the three locations, there is a general decrease in LCOE from C-I to C-III and a slight increase going from C-III to C-IV in Nigeria and Zambia; for the case of Ethiopia, the increase in LCOE in C-IV is significantly high resulting in values greater than those of C-I minigrid model. The results highlight the importance of low technology cost, low cost of capital and productive loads in minigrid energy cost. The increase in LCOE in case C-IV partly due to low load factor of the increasing seasonal irrigation loads underscores the need for consideration of other PUE loads, such as post harvest processing (Falchetta et al., 2021; Farthing et al., 2023) beyond primary irrigation loads, to further sustain the decrease in LCOE with agricultural loads.

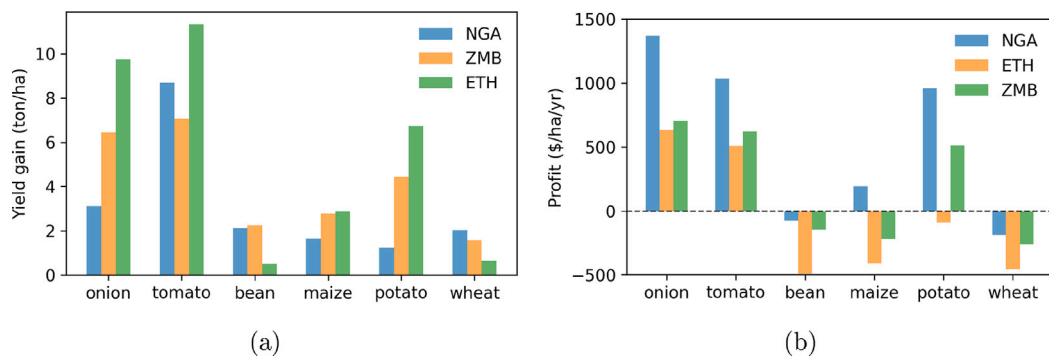


Fig. 4. (a) Increase in crop yield due to irrigation and; (b) Annual profitability of irrigation in the three location sites in Zambia (ZMB), Ethiopia (ETH), and Nigeria (NGA) based on C-II minigrid LCOE in Fig. 2 as irrigation energy tariffs.

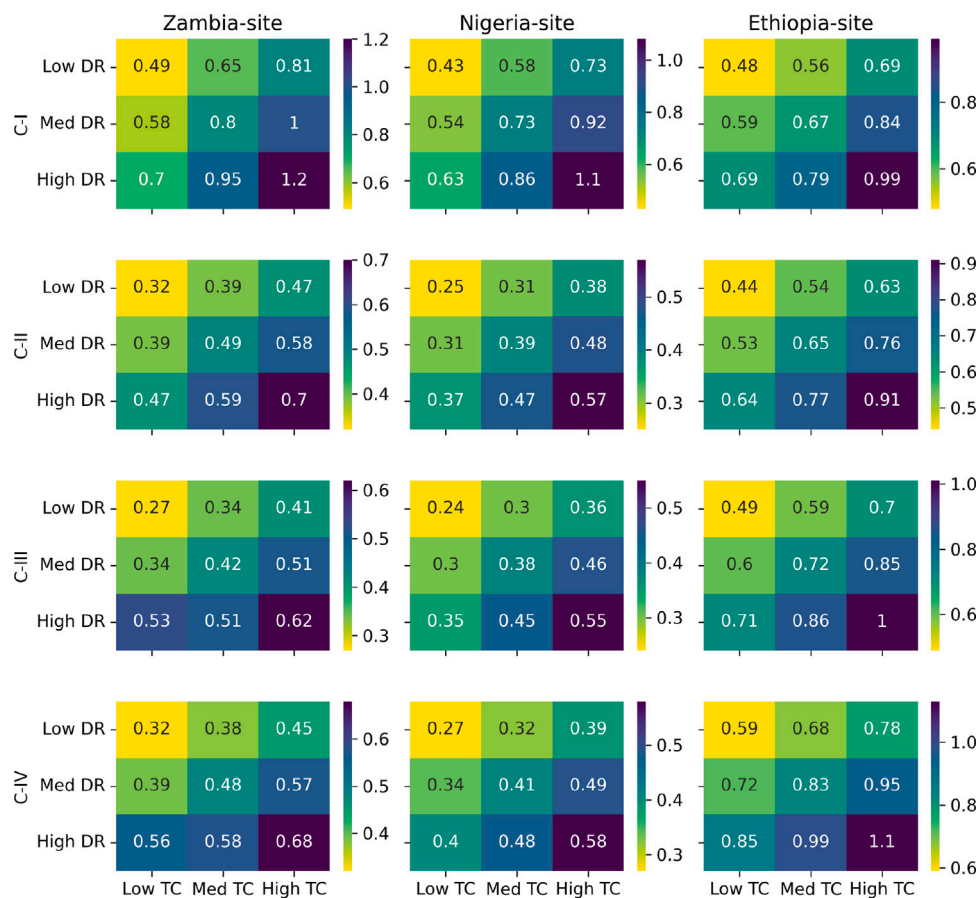


Fig. 5. Levelised cost of energy (LCOE) of the minigrid with different cost of capital (DR) and technology cost (TC) scenarios for the four cases (C-I, C-II, C-III, and C-IV).

Irrigation profitability

In this sub-section, we present sensitivity analysis of irrigation profitability with changes in system costs; that is the low-cost, median-cost, and high-cost scenarios described in Section “Minigrid Levelized Cost of Energy”. For irrigation energy cost, uncertainties in cost drivers, both DR and TC, are factored in the sensitivity analyses of LCOE, results in Fig. 5. In Fig. 6, we present sensitivity results of irrigation profitability with different cost scenarios and soil fertility levels. Besides crops and location, irrigation profitability decreases with a decrease in soil fertility and change from low-cost to high-cost scenarios. The highest profits are realised by the combination of low-cost scenario and optimal (100f) soil fertility rates, with onion crop reaching \$ 1600/ha/year under a combination of ‘low-cost’ and 100f soil fertility scenarios. It is generally not viable to irrigate at 25f soil fertility with the exception of onion, tomato and potato crops in Nigeria. Generally, low-value

crops (mainly maize, beans, and wheat) are less profitable to irrigate under all scenarios considered in this paper. This highlights the need to explore further cost-cutting measures to understand conditions under which it might be viable to irrigate these crops in the nexus planning of minigrids with smallholder irrigation.

In Fig. 7, we present the results of an innovative business model designed to armotise different cost components of the irrigation and water access infrastructure. The results highlight the benefits of lowering upfront cost barriers to irrigation adoption; they do not only enable irrigation adoption but also enhance profitability. The highest profits are realised when all upfront costs are armotised by the innovative PAYGo business model. On the other hand, ‘zero incentive’ model is the least profitable model. An important observation to highlight is that economic viability of less profitable crops such as beans, and maize is enhanced by the proposed incentive model; as shown in the figure,

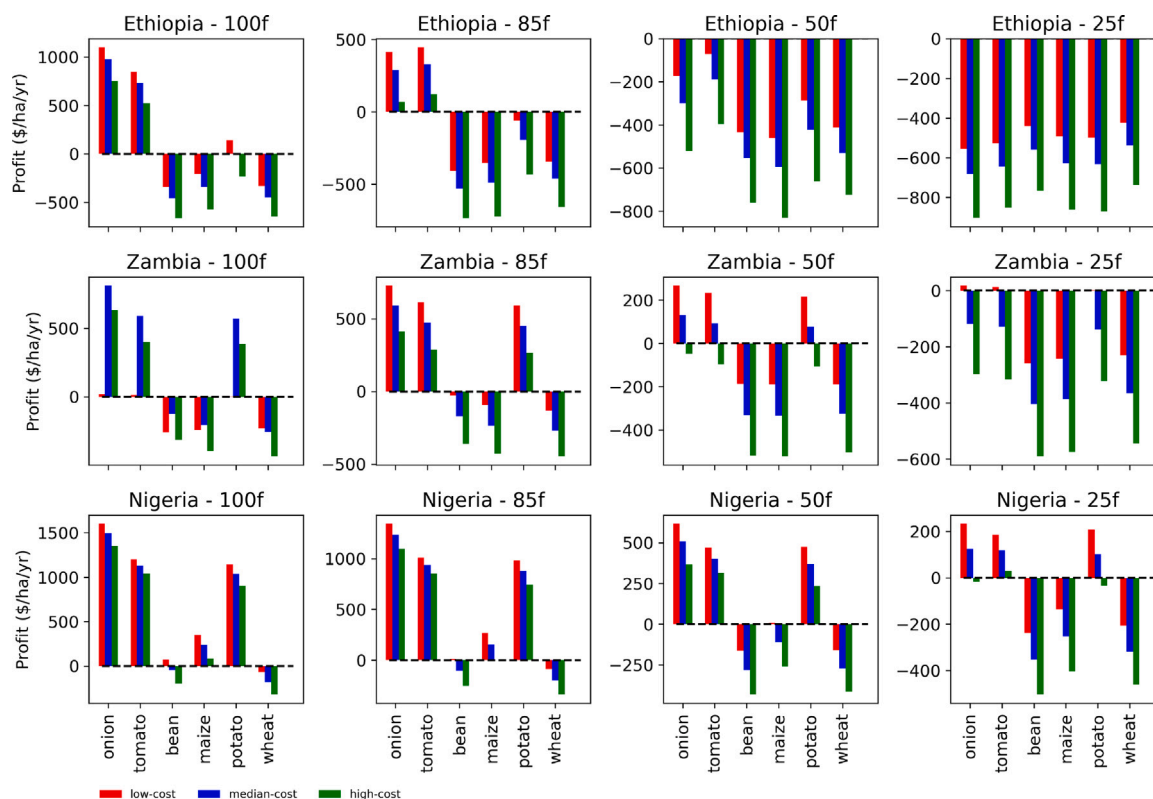


Fig. 6. Sensitivity analysis of different cost scenarios and soil fertility conditions on irrigation profitability. Note that each sub-figure in the figure has its own y-axis scale.

profitability of maize crop in Nigeria increased from ‘Zero incentive’ to ‘All costs incentive’. The findings are consistent with prior works in the literature (Falchetta et al., 2023); that alleviating upfront cost barriers to irrigation adoption enhances irrigation viability.

A general observation from the presented sensitivity analyses results is that horticultural crops seem by far the most profitable in terms of irrigation input benefit, while profitability for staple crops is more limited and depends on other factors (such as irrigation costs, geographical location, and crop prices). This is an important result from a policy point of view, and also worth remarking that often horticultural crops necessitates the need for cold storage (an additional electrical load which could be met by a minigrid) for products to be preserved and subsequently sold, i.e. to avoid food (and therefore income) loss. Potential energy demand for cold storage – not analysed in this paper – can mitigate seasonality of irrigation loads, enhance the load factor with the resultant benefits of reduced minigrid LCOE.

Policy implications

We have shown from our model results the potential benefits of irrigation loads in reducing minigrid LCOE, which is consistent with previous works in the literature (Shoeb & Shafiullah, 2018; Uamusse et al., 2019). We have also shown that irrigation profitability is a function of a complex interplay of multiple factors including the type of crops grown, fertiliser application rate, the irrigation infrastructure costs, irrigation energy prices, and farm-gate crop prices—which are exogenous to the model. We have further shown the potential of innovative PAYGo business model incentives in alleviating irrigation adoption barrier and enhancing irrigation profitability.

However, it is worth remarking here that the proposed minigrid-irrigation nexus is complex and its implementation and exploitation of the benefits thereof will require supportive policy environment for both the farmer and the minigrid operator. It is acknowledged in the current literature that lack of supportive policies is one of the sources of risks that discourage private sector investment in the minigrid sector

in SSA (Falchetta et al., 2022; Peters et al., 2019) with potential to cause project failures (Ikejamba, Mpuan, Schuur, & Van Hillegersberg, 2017). In this regard, we propose the following policy considerations for planning at the nexus of minigrid electrification and smallholder irrigation:

First, we emphasise the need for an integrated system approach with synergistic partnerships of key actors, including public institutions that handle agricultural modernisation and energy access functions, policy makers, minigrid developers, financiers, and development partners and donors. For smallholder farmers, such a partnership should aim at enhancing access to farm inputs (seeds and fertilisers), credit for irrigation infrastructure development, access to crop markets, and access to agricultural extension services to unlock agricultural potential (Jama & Pizarro, 2008). It is estimated that only about 50% of the farmers in SSA use inorganic fertilisers, while 10% do not apply any artificial fertiliser due to their high costs (Vanlauwe et al., 2014). Our results show that it is not economically viable to irrigate at a very low fertiliser application rate (25f in this case—a scenario that models non-application of artificial fertilisers). Therefore, we argue here that access to the right quantity of fertilisers is cardinal for high agricultural yield (Jama & Pizarro, 2008) and profitable deployment of the nexus project as demonstrated by our results in Fig. 6.

Secondly, for the minigrid developers, the nexus partnership should aim at mitigating regulatory and financial risks (Agenbroad et al., 2018; Falchetta et al., 2022). Access to low cost of capital and low cost technology is still a critical challenge in SSA, particularly when one considers the nascency of off-grid electrification (Peters et al., 2019). Where financing is accessible, lending rates are high, often more than 15% in many countries (Agenbroad et al., 2018) (which is the case for the results in this paper as shown in Table SI 11 in the supplement). The implications of this high interest rates is the high minigrid project costs to the developer and the end-users (Falchetta, 2021). Mitigation of economic and financial risks—mainly those related to access to finance—can reduce overall system costs and encourage investment in the minigrid sector. Our results in Fig. 5 shows that, on

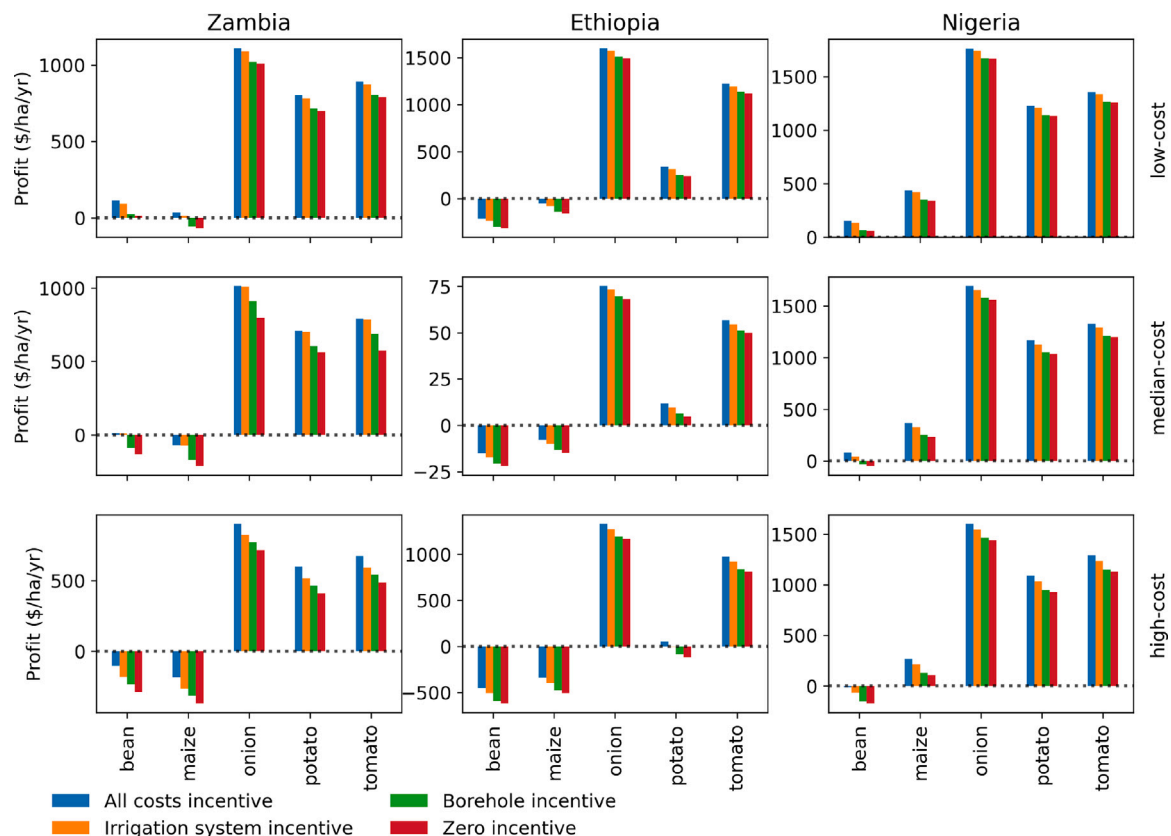


Fig. 7. Sensitivity analysis on different PAYGo model incentives for the capital costs of irrigation. The results are based on optimal soil fertility (100f) assumptions. The negative profit values are scenarios where the capital cost of the system cannot be recovered within the project life of 20 years. Note that wheat crop is excluded in the analysis because it is unprofitable under all scenarios and business incentives.

average, a reduction of the cost of capital by 5% reduces the LCOE of the baseline minigrid (the minigrid without irrigation loads) by \$0.14/kWh, which is equivalent to 19% decrease.

Thirdly, we highlight the benefits of PAYGo business model incentives designed to alleviate upfront capital cost barriers to the adoption of irrigation. Despite demonstrated economic benefits of irrigation, many smallholders in SSA cannot afford the high capital costs of irrigation infrastructure (Lebdi, 2016) and therefore financial incentives designed to amortise all or a part of the upfront capital costs are critical in unlocking uptake and profitable operation of minigrid-powered smallholder irrigation in SSA.

Lastly, our results shows that horticultural crops are by far the most profitable to irrigate while irrigation profitability of staple crops is dependent of other factors (such as costs and location). It is worth remarking here that horticultural crops often necessitates the need for cold storage (which has further cooling electricity demand) for the crop proceeds to be sold at a later date to minimise crop and income loss. Cold storage has potential to increase the farmers' income through market price stabilisation as well as reducing post-harvest waste, two major problems in SSA. Cold storage energy demand from the minigrid has potential to mitigate the effects of seasonality of irrigation loads of the utilisation factor of the minigrid assets resulting in LCOE reduction. This underscores that horticultural crops are the most favourable crops for the nexus deployment of minigrids with irrigation loads.

Conclusion

In our study, we investigated the impacts of irrigation loads on the levelised cost of minigrid energy (LCOE) as well as the appropriate conditions under which both the minigrid developer and smallholder

farmers can realise positive returns when they operate together. The results of our study show that the change in LCOE of minigrid with irrigation loads depend on the quantity of irrigable land and its spatial location. Irrigation energy demand for a 30 ha of farmland, in case C-II minigrid, results in up to 47% decrease in LCOE, with a further increase in irrigable land resulting in increase in the minigrid LCOE across the case study sites. A sensitivity analysis of LCOE with uncertainties in the cost of capital (discount rates) and technology costs (minigrid equipment costs) show that low-cost scenario (low cost of capital and low technology cost) guarantees LCOE values of less than \$0.60/kWh in all the three case study locations.

On the part of the smallholder farmer, sensitivity analysis of the cost of capital and technology costs on irrigation profitability shows strong dependence of economic viability of irrigation on the crop grown, fertiliser application rates, geographical location, irrigation infrastructure costs, irrigation energy cost, and, to some extent, farm-gate crop prices. In descending order, onion, tomato, and potato are the most viable crops to irrigate at the obtained minigrid energy cost, particularly considering the applications of fertilisers that are near optimal to optimal (85f–100f). Low-value crops (common staples—maize, wheat, bean crops) record low economic viability of irrigation; wheat crop is not viable to irrigate while beans and maize record low profitability under low-cost production scenario. We observed a decrease in irrigation profitability with a decrease in soil fertility levels, which is consistent with the results in Izar-Tenorio et al. (2021). We further assess the effects of innovative pay-as-you-go (PAYGo) business model incentives to irrigation profitability; our results show positive benefits with the highest profitability recorded for the scenario that amortises all the cost of irrigation infrastructure. Besides increasing profitability of the already profitable crops (onion, tomato and potato), PAYGo incentives

also enhances economic viability of less profitable crops. Maize and bean crops are economically viable to irrigate under PAYGo business models and cost scenarios in some geographical locations but wheat crop is generally not viable to irrigate under conditions considered in this paper.

In conclusion, our model results underscore that profitable deployment of minigrids with smallholder irrigation depend on the geographical location, the choice of crop to grow, fertiliser application rate, cost of capital and technology costs. Horticultural crops under low-cost scenario and optimal fertiliser application rates are the most suitable conditions for the smallholder farmer to operate profitably in a joint operation with minigrids. Our model and results thereof can be used as high-level decision references when planning joint deployment of minigrids with irrigation loads.

To advance our work, future research will explore holistic assessment of irrigation and post-harvest energy demand on the minigrid operation to study the potential benefits of cold storage to further reduce the minigrid LCOE. Another potential study area is large scale national and continental scale analysis of irrigation potential in Sub-Saharan Africa. Such a work may include quantification of irrigation profitability of the most important crops required to meet the region's fast growing population's food needs. Also important is the assessment of irrigation energy and infrastructure capacity requirement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.esd.2023.101345>.

References

- Agnew, J., Carlin, K., Ernst, K., & Doig, S. (2018). *Minigrids in the money: Six ways to reduce minigrid costs by 60% for rural electrification*. Tech. Rep., Rocky Mountain Institute, [Online]. Available: www.rmi.org/insight/minigrids2018.
- Allan, R., Pereira, L., & Smith, M. (1998). *Crop evapotranspiration-Guidelines for computing crop water requirements-FAO Irrigation and drainage paper 56, Vol. 56*. Food and Agricultural Organisation.
- Alliance for Green Revolution in Africa (AGRA) (2014). *Africa agriculture status report 2014: Climate change and smallholder agriculture in Sub-Saharan Africa*. Alliance for Green Revolution in Africa (AGRA, ch. Section II: Agriculture Data for selected Sub-Saharan African Countries.
- Ba, M. N. (2017). Competitiveness of maize value chains for smallholders in West Africa: Case of Benin, Ghana and Cote D'Ivoire. *Agricultural Sciences*, 8(12), 1372–1401.
- Babovic, J., Milic, S., & Radojevic, V. (2009). Economics effects of irrigation in plant production. *Economics of Agriculture*, 56(297-2016-3734), 41–53.
- Bhattacharyya, S. C. (2013). Financing energy access and off-grid electrification: A review of status, options and challenges. *Renewable and Sustainable Energy Reviews*, 20, 462–472.
- Blodgett, C., Moder, E., Kickham, L., & Leaf, H. (2016). *Powering productivity: Early insights into mini grid operations in rural Kenya*. Tech. Rep., Vulcan Impact Investing.
- Blum, N. U., Wakeling, R. S., & Schmidt, T. S. (2013). Rural electrification through village grids—Assessing the cost competitiveness of isolated renewable energy technologies in Indonesia. *Renewable and Sustainable Energy Reviews*, 22, 482–496.
- Bonan, J., Pareglio, S., & Tavoni, M. (2017). Access to modern energy: a review of barriers, drivers and impacts. *Environment and Development Economics*, 22(5), 491–516.
- Booth, S., Li, X., Baring-Gould, I., Kollanyi, D., Bharadwaj, A., & Weston, P. (2018). *Productive use of energy in african micro-grids: Technical and business considerations*. Tech. Rep., Golden, CO (United States): National Renewable Energy Lab.(NREL).
- Burney, J. A., & Naylor, R. L. (2012). Smallholder irrigation as a poverty alleviation tool in sub-Saharan Africa. *World Development*, 40(1), 110–123.
- Burney, J. A., Naylor, R. L., & Postel, S. L. (2013). The case for distributed irrigation as a development priority in sub-Saharan Africa. *Proceedings of the National Academy of Sciences*, 110(31), 12513–12517.
- Cabraal, R. A., Barnes, D. F., & Agarwal, S. G. (2005). Productive uses of energy for rural development. *Annual Review of Environment and Resources*, 30, 117–144.
- Center for International Earth Science Information Network - CIESIN - Columbia University (2016). Gridded population of the world, version 4 (GPWv4): Population count. <https://sedac.ciesin.columbia.edu/data/collection/gpw-v4>, Accessed: 2023-10-08.
- Christiansen, L., Demery, L., & Kuhl, J. (2011). The (evolving) role of agriculture in poverty reduction—An empirical perspective. *Journal of Development Economics*, 96(2), 239–254.
- Ciller, P., Lumberras, S., & González-García, A. (2021). Network cost estimation for mini-grids in large-scale rural electrification planning. *Energies*, 14(21), 7382.
- Dagnachew, A. G., Choi, S.-M., & Falchetta, G. (2023). Energy planning in Sub-Saharan African countries needs to explicitly consider productive uses of electricity. *Scientific Reports*, 13(1).
- Daly, H., & Walton, M. (2017). *Energy access outlook: from poverty to prosperity, world energy outlook-2017 special report*. Tech. Rep., International Energy Agency, [Online]. Available: https://read.oecd-ilibrary.org/energy/energy-access-outlook-2017_9789264285569-en#page1.
- El-Shimy, M. (2013). Sizing optimisation of stand-alone photovoltaic generators for irrigation water pumping systems. *International Journal of Sustainable Energy*, 32(5), 333–350.
- ENEA Consulting (2020). *Scaling up minigrids for rural electrification: Lessons from AECF's project portfolio*. Tech. Rep., Enea Consulting, [Online]. Available: <https://www.aecfafrica.org/wp-content/uploads/2021/12/AECF-Scaling-up-mini-grids-18-Nov-2020.pdf>.
- Energy Market Accelerator Ethiopia (AMAE) (2021). Ethiopia off-grid energy portal-decentralised renewable energy and modalities (DREAM) project launched in Ethiopia. [Online]. Available: <https://www.etoffgrid.org/2021/10/13/decentralized-renewable-energy-agriculture-modalities-dream-launched-in-ethiopia/>.
- Energy Sector Management Assistance Program (ESMAP) (2019). *Mini grids for half a billion people: Market outlook and handbook for decision makers*. Tech. Rep., The World Bank's Energy Sector Management Assistance Program, [Online]. Available: <https://openknowledge.worldbank.org/bitstreams/f3815820-92b6-5807-8e9f-d0bd98732b5a/download>.
- Energy Sector Management Assistance Program (ESMAP) (2022). *Minigrids for half a billion people: Market outlook and handbook for decision makers*. Tech. Rep., Energy Sector Management Assistance Program, [Online]. Available: https://www.esmap.org/Mini_Grids_for_Half_a_Billion_People_The_Report.
- Falchetta, G. (2021). Energy access investment, agricultural profitability, and rural development: time for an integrated approach. *Environmental Research: Infrastructure and Sustainability*.
- Falchetta, G., Michoud, B., Hafner, M., & Rother, M. (2022). Harnessing finance for a new era of decentralised electricity access: A review of private investment patterns and emerging business models. *Energy Research & Social Science*, 90.
- Falchetta, G., Semeria, F., Tuninetti, M., Giordano, V., Pachauri, S., & Byers, E. (2023). Solar irrigation in sub-Saharan Africa: economic feasibility and development potential. *Environmental Research Letters*, 18(9).
- Falchetta, G., Stevanato, N., Moner-Girona, M., Mazzoni, D., Colombo, E., & Hafner, M. (2021). The M-LED platform: advancing electricity demand assessment for communities living in energy poverty. *Environmental Research Letters*, 16(7), Article 074038.
- Farthing, A., Rosenlieb, E., Steward, D., Reber, T., Njobvu, C., & Moyo, C. (2023). Quantifying agricultural productive use of energy load in Sub-Saharan Africa and its impact on microgrid configurations and costs. *Applied Energy*, 343.
- Food and Agricultural Organisation (FAO) (2022). Crop production, yield, harvested area and processed (global - national - annual - FAOSTAT). <https://data.apps.fao.org/catalog/dataset/crop-production-yield-harvested-area-global-national-annual-faostat>, Accessed: 2023-10-18.
- Fuglie, K., & Rada, N. (2013). *Resources, policies, and agricultural productivity in sub-Saharan Africa: USDA-ERS Economic Research Report, no. 145*.
- Giller, K. E., Delaune, T., Silva, J. V., van Wijk, M., Hammond, J., Descheemaeker, K., et al. (2021). Small farms and development in sub-Saharan Africa: Farming for food, for income or for lack of better options? *Food Security*, 13(6), 1431–1454.
- Hagos, F., Makombe, G., Namara, R. E., & Awulachew, S. B. (2009). Importance of irrigated agriculture to the Ethiopian economy: Capturing the direct net benefits of irrigation. *Vol. 128. IWMI*.
- Holzziel, J. (2021). *Success in rural electrification framework case studies: Ethiopia, integrating off-grid technologies into electrification planning*. Tech. Rep., Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH, [Online]. Available: <https://africa-energy-portal.org/sites/default/files/2022-03/Success-in-Rural-Electrification-Case-Study-Ethiopia.pdf>.
- Ikejamba, E. C., Mpuan, P. B., Schuur, P. C., & Van Hillegersberg, J. (2017). The empirical reality & sustainable management failures of renewable energy projects in Sub-Saharan Africa (part 1 of 2). *Renewable Energy*, 102, 234–240.

- International Energy Agency (2022). *World energy outlook special report: Africa energy outlook 2022: Tech. Rep.*, IEA: International Energy Agency, [Online]. Available: <https://www.iea.org/reports/africa-energy-outlook-2022/executive-summary>.
- International Finance Corporation (IFC) (2018). *Benchmarking minigrid DESCOS, 2017 update: Summary findings: Tech. Rep.*, International Finance Corporation, [Online]. Available: https://gallery.mailchimp.com/39ef5953cb296ad051b58bd59/files/59c4d023-2094-49d8-bc71-9f96b9f5188f/20180516_Benchmarking_2.0_Final_to_Share.pdf.
- International Food Policy Research Institute (2020). Spatially-disaggregated crop production statistics data in Africa South of the Sahara for 2017. <http://dx.doi.org/10.7910/DVN/FSSKBW>, Accessed: 2023-08-28.
- International Renewable Energy Agency (IRENA) (2016). *Solar pumping for irrigation: Improving livelihoods and sustainability: Tech. Rep.*, The International Renewable Energy Agency (IRENA), [Online]. Available: <https://www.irena.org/publications/2016/Jun/Solar-Pumping-for-Irrigation-Improving-livelihoods-and-sustainability>.
- Izar-Tenorio, J. L., Jaramillo, P., & Williams, N. (2021). Techno-economic feasibility of small-scale pressurized irrigation in Ethiopia, Rwanda, and Uganda through an integrated modeling approach. *Environmental Research Letters*, 16(10), Article 104048.
- Jama, B., & Pizarro, G. (2008). Agriculture in Africa: Strategies to improve and sustain smallholder production systems. *Annals of the New York Academy of Sciences*, 1136(1), 218–232.
- Jayne, T. S., Mather, D., & Mghenyi, E. (2010). Principal challenges confronting smallholder agriculture in sub-Saharan Africa. *World Development*, 38(10), 1384–1398.
- Juturu, P., Conlon, T., Zhang, Y., Avraam, C., Siddiqui, S., Simane, B., et al. (2023). Optimal grid expansion under future electricity demand for groundwater irrigation in Ethiopia. *Energy for Sustainable Development*, 72, 351–377.
- Kumar, M. D. (2005). Impact of electricity prices and volumetric water allocation on energy and groundwater demand management: Analysis from Western India. *Energy Policy*, 33(1), 39–51.
- Kyriakarakos, G., Balaoutis, A. T., & Bochtis, D. (2020). Proposing a paradigm shift in rural electrification investments in Sub-Saharan Africa through Agriculture. *Sustainability*, 12(8), 3096.
- Lebdi, F. (2016). *Irrigation for agricultural transformation: Tech. Rep.*, Africa Center for Economic Transformation and Japan International Cooperation Agency Research institute (JICA-RI), [Online]. Available: <https://www.jica.go.jp/jica-ri/publication/booksandreports/175nbg0000004aet-att/175nbg0000004ai9.pdf>.
- Li, X., Salasovich, J., & Reber, T. (2018). Microgrid load and LCOE modelling results. [Online]. Available: <https://data.nrel.gov/submissions/79>.
- MacDonald, A. M., Bonsor, H. C., Dochartaigh, B. É. Ó., & Taylor, R. G. (2012). Quantitative maps of groundwater resources in Africa. *Environmental Research Letters*, 7(2).
- Maqelepo, L., Williams, N., & Taneja, J. (2022). Rural electrification subsidy estimation: a spatial model development and case study. *Environmental Research: Infrastructure and Sustainability*, 2(4), Article 045009.
- McCall, M., & Santana, S. (2018). *Closing the circuit: Stimulating end-use demand for rural electrification: Tech. Rep.*, Rocky Mountain Institute, [Online]. Available: <https://rmi.org/wp-content/uploads/2018/10/SEED-Demand-Stimulation-Report.pdf>.
- Mentis, D., Howells, M., Rogner, H., Korkovelos, A., Arderne, C., Zepeda, E., et al. (2017). Lighting the World: the first application of an open source, spatial electrification tool (OnSSET) on Sub-Saharan Africa. *Environmental Research Letters*, 12(8).
- Mergulhao, V. P., Capra, L., Voglitsis, K., & Parikh, P. (2023). How do they pay as they go?: Learning payment patterns from solar home system users data in Rwanda and Kenya. *Energy for Sustainable Development*, 76.
- Ngowi, J. M., Bångens, L., & Ahlgren, E. O. (2019). Benefits and challenges to productive use of off-grid rural electrification: The case of mini-hydropower in Bulongwa-Tanzania. *Energy for Sustainable Development*, 53, 97–103.
- Nilsson, A., Mentis, D., Korkovelos, A., & Otmani, J. (2021). A GIS-based approach to estimate electricity requirements for small-scale groundwater irrigation. *ISPRS International Journal of Geo-Information*, 10(11).
- Peters, J., Sievert, M., & Toman, M. A. (2019). Rural electrification through mini-grids: Challenges ahead. *Energy Policy*, 132, 27–31.
- Phocaides, A. (2007). *Handbook on pressurized irrigation techniques*. Food & Agriculture Org..
- Reckefeller Foundation (2015). Perspectives to reducing post-harvest losses of agricultural products in africa. African Development Bank Group, https://www.afdb.org/fileadmin/uploads/afdb/Documents/Events/DakAgri2015/Agriculture_Industrialization_and_post-harvest_losses.pdf.
- Shah, T., Namara, R., & Rajan, A. (2020). *Accelerating irrigation expansion in Sub-Saharan Africa: policy lessons from the global revolution in farmer-led smallholder irrigation*. IWMI.
- Shah, T., Verma, S., & Pavelic, P. (2013). Understanding smallholder irrigation in Sub-Saharan Africa: results of a sample survey from nine countries. *Water International*, 38(6), 809–826.
- Shoeb, M. A., & Shafiullah, G. (2018). Renewable energy integrated islanded microgrid for sustainable irrigation—A Bangladesh perspective. *Energies*, 11(5).
- Stackhouse, P. (2022). NASA power | prediction of worldwide energy. [Online]. Available: <https://power.larc.nasa.gov/>.
- Suryani, A., Fadhilah, A., Saichu, P., & Mubarak, H. (2021). *PV minigrid installation: Dos and don'ts: Tech. Rep.*, Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH, [Online]. Available: <https://endev.info/wp-content/uploads/2021/06/PV-Mini-Grid-Dos-and-Donts.pdf>.
- The World Bank Group (2018). Africa electricity grids explorer. [Online]. Available: <https://africagrid.energydata.info/>.
- Uamusse, M. M., Tussupova, K., Persson, K. M., & Berndtsson, R. (2019). Mini-grid hydropower for rural electrification in mozambique: Meeting local needs with supply in a nexus approach. *Water*, 11(2).
- (2020). *World cities report 2020: The value of sustainable urbanization: Tech. Rep.*, UN-Habitat, [Online]. Available: https://unhabitat.org/sites/default/files/2020/10/wcr_2020_report.pdf.
- Vanlauwe, B., Coyne, D., Gockowski, J., Hauser, S., Huising, J., Masso, C., et al. (2014). Sustainable intensification and the African smallholder farmer. *Current Opinion in Environmental Sustainability*, 8, 15–22.
- Vanuytrecht, E., Raes, D., Steduto, P., Hsiao, T. C., Fereres, E., Heng, L. K., et al. (2014). AquaCrop: FAO's crop water productivity and yield response model. *Environmental Modelling & Software*, 62, 351–360.
- Veritas Consulting (2020). *Investment report for a minigrid pilot program at agriculture commercial clusters in Ethiopia: Report*, Federal Democratic Republic of Ethiopia.
- Villholth, K. G. (2013). Groundwater irrigation for smallholders in Sub-Saharan Africa—a synthesis of current knowledge to guide sustainable outcomes. *Water International*, 38(4), 369–391.
- Wamalwa, F., Izar-Tenorio, J. L., & Williams, N. (2022). Spatial assessment of solar PV-based irrigation potential in Kenya. In *2022 IEEE PES/IAS powerAfrica* (pp. 1–5). IEEE.
- Williams, N. J., Jaramillo, P., & Taneja, J. (2018). An investment risk assessment of microgrid utilities for rural electrification using the stochastic techno-economic microgrid model: A case study in Rwanda. *Energy for Sustainable Development*, 42, 87–96.
- Xie, H., Ringler, C., & Mondal, M. A. H. (2021). Solar or diesel: A comparison of costs for groundwater-fed irrigation in sub-saharan africa under two energy solutions. *Earth's Future*, 9(4).
- You, L., Wood, S., & Wood-Sichra, U. (2009). Generating plausible crop distribution maps for Sub-Saharan Africa using a spatially disaggregated data fusion and optimization approach. *Agricultural Systems*, 99(2–3), 126–140.